



Batterjee Medical College
For Science and Technology

Cell membrane and its functions

Dr. Hader I. Sakr

Associate Professor, Medical Physiology

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

﴿وَاتَّقُوا اللَّهَ ۖ وَيَعْلَمَ كُمْ اللَّهُ ۖ وَاللَّهُ

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Learning outcomes:

By the end of the lecture, the student will be able to:

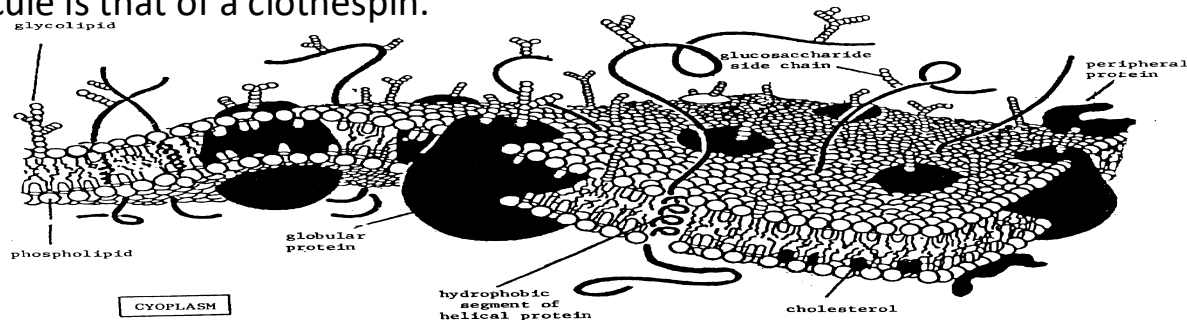
- **Describe the composition of the cell membrane**
- **Outline the different functions of the cell membrane**
- **Explain the different methods of transport across the cell membrane**



The cell membrane and its functions

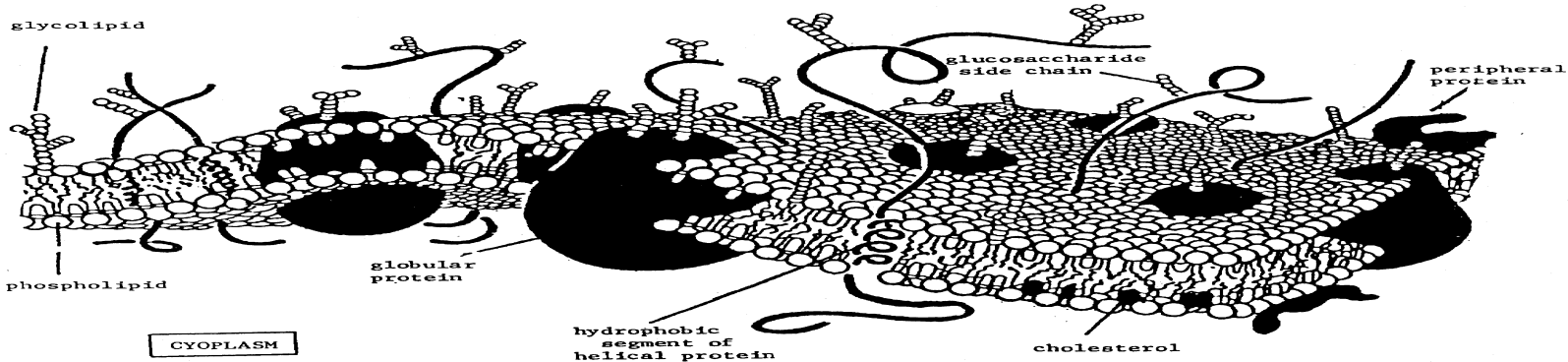
The cell membrane and its functions

- The CM, that surrounds the cell completely, is a very thin elastic and **semipermeable** membrane of about 7.5-10 nm thick (75 -100 Angstrom units = $\text{\AA} = 10^{-10}$ meter).
- It is composed mainly of proteins and lipids; the approximate composition is:
 - Proteins **55%**
 - Phospholipids (PLs) Lipids **25 %**
 - Cholesterol **13%**
 - Other lipids **4%**
 - Carbohydrates **3%**
- The shape of PLs molecule is that of a clothespin.



The cell membrane and its functions

- A head which contains the **phosphate** radical (water soluble, **hydrophilic**, **polar**) exposed to the aqueous environment that bathes the exterior of the cells and the aqueous cytoplasm.
- Two tails that contain the **fatty acids** radicals which are relatively **insoluble in water** (**hydrophobic**, **non-polar**) meet in the water poor interior of the membrane.
- Thus, the molecule is **amphipathic**: that is; it has one part **hydrophilic** and the other **hydrophobic**.



The cell membrane and its functions

1. The lipid bilayer is important for the **flexibility** and selective **permeability**.
2. The cholesterol molecules affect membrane **permeability** and give it **toughness**.
3. The cell membrane proteins are present floating in the lipid bilayer. ***Two types of proteins:***
 - **The integral proteins.**
 - **The peripheral proteins.**
4. The membrane carbohydrates occur almost in combinations with proteins and lipids in the form of **glycoproteins** and **glycolipids**. They are important in:

1. Acting as recognition sites.

2. Self-differentiation.

3. Immune responses.

4. Helping attaching the cells together.



Functions of cell membrane proteins

Functions of cell membrane proteins

1. Structural proteins: They keep the integrity of the membrane and give it strength.

2. Form passive channels:

3. Carriers in facilitated diffusion:

4. Carriers in active transport (=Pumps):

integral

5. Receptors:

peripheral

6. Enzymes:

7. Identity proteins: They are mostly glycoproteins which give the cells the individual's label of

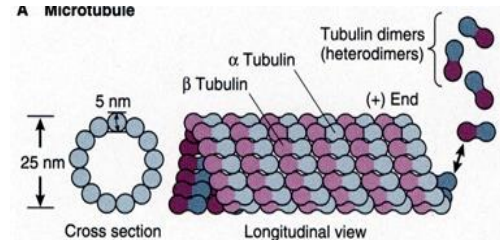
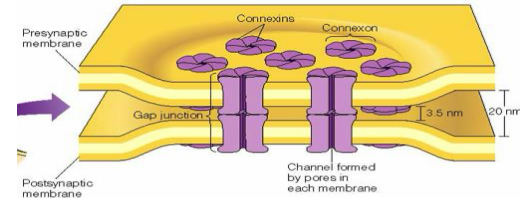
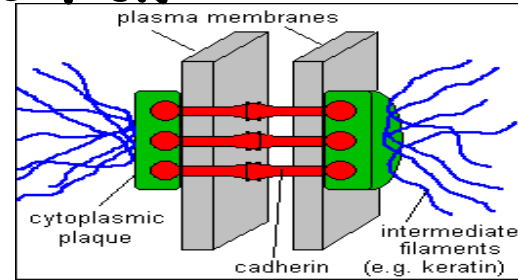
identity, so that they would not be attacked by the immune system. → glycoprotein

Functions of cell membrane proteins

8. Intercellular connections:

8 and 9 connection

- Fasten the cells to one another & to the surrounding tissues.
 1. Tight junctions (zonula occludence).
 2. Desmosomes.
 3. Zonula adherence.
- Permit transfer of ions & other molecules from one cell to another:
 1. Gap junctions.



9. Cell adhesion molecules (CAMs): Attach cell to basal lamina.

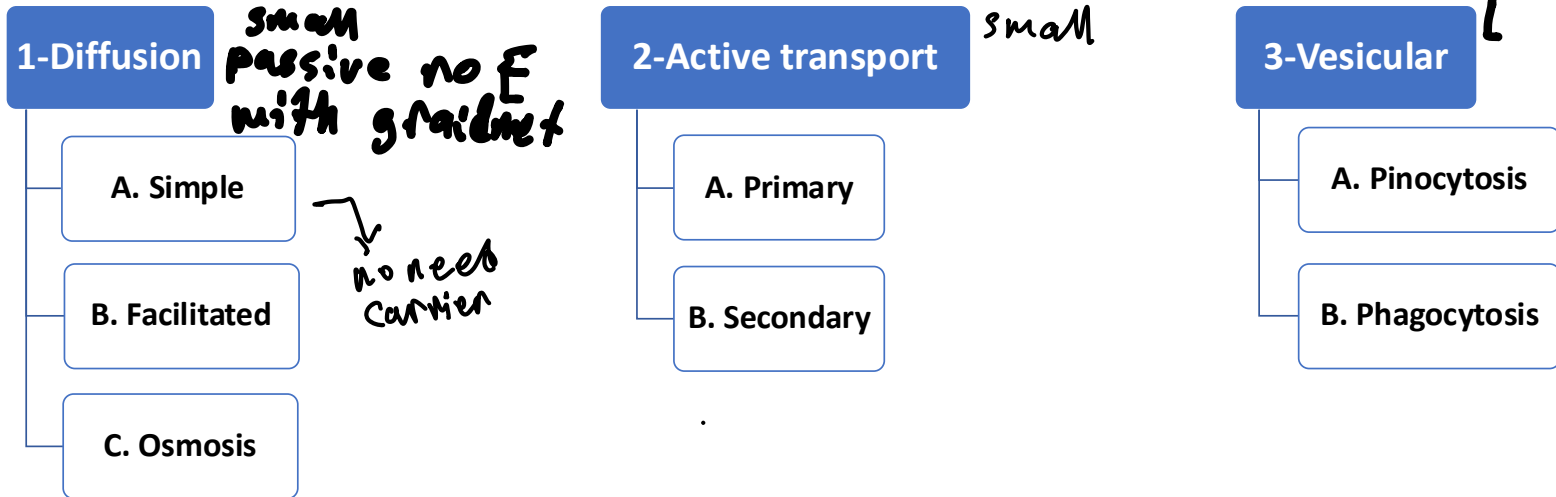
10. Cytoskeleton of the cell: System of fibers that maintains the structure of the cell and permits it to change shape and move.



Different methods of transport across the cell membrane

Different methods of transport across the cell membrane

- The differences between the components of ICF and ECF are extremely important to the life of the cell. The mechanisms that control the transport through the cell membranes are so important to maintain these differences. Substances can pass through a cell membrane in three separate ways:



Different methods of transport across the cell membrane

1. Diffusion.

A. Simple Diffusion.

- It means that the molecules or ions will **diffuse** through the membrane, **according** to the concentration **gradient**, **passively without** binding with **carrier proteins** in the membrane. The rate of diffusion is affected by many factors that are summarized in the formula.

Diffusion rate $\propto$ $\frac{\text{Conc. Difference} \times \text{Surface area of the membrane} \times \text{Temp.}}{\text{Distance} \times \sqrt{\text{Molecular Weight}}}$

direct $\nearrow$

inverse $\nwarrow$

☆

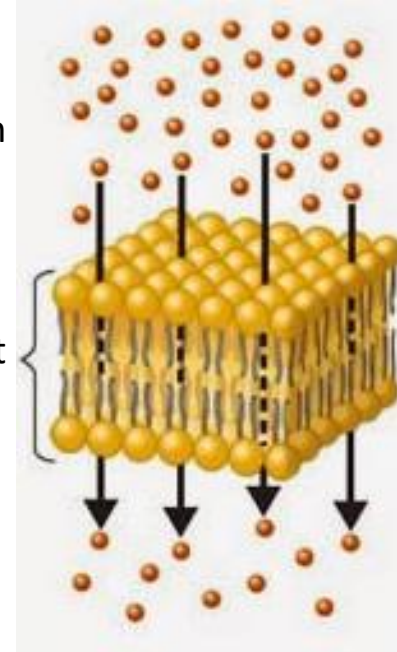
Different methods of transport across the cell membrane

➤ Simple diffusion can occur across the CM through the **lipid bilayer** or **protein channels**:

1. Through the lipid bilayer.

a) The lipid soluble substances: such as **oxygen** and **nitrogen** can dissolve directly in lipid bilayer through CM.

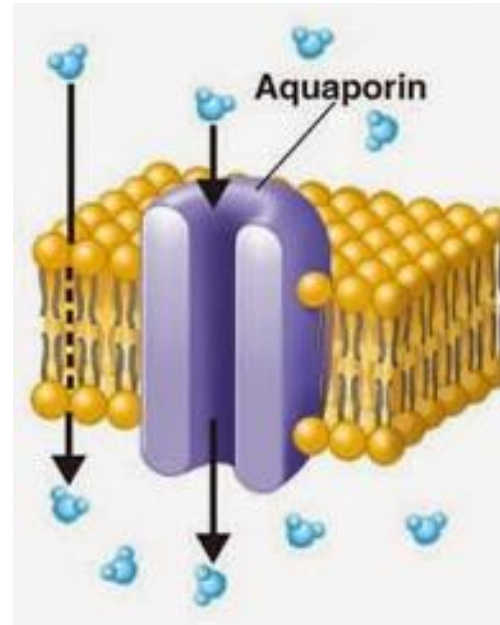
- The rate of diffusion is directly proportional to their lipid solubility.
- O_2 can be delivered to the cell interior almost as though CM did not exist.



Different methods of transport across the cell membrane

b) Water molecules:

- They pass **directly** through the lipid bilayer and also through protein channels (**aquaporin**) as well.
- Water is highly **insoluble** in the membrane lipids; but as water molecules are **small enough** and their **kinetic energy is great** enough that they can simply **penetrate like bullets** through the lipid portion of the membrane before the hydrophobic character of the lipids can stop them.



- c) Lipid insoluble molecules can pass through the lipid bilayer in the same way as water molecules if they are **small enough** and **uncharged**.

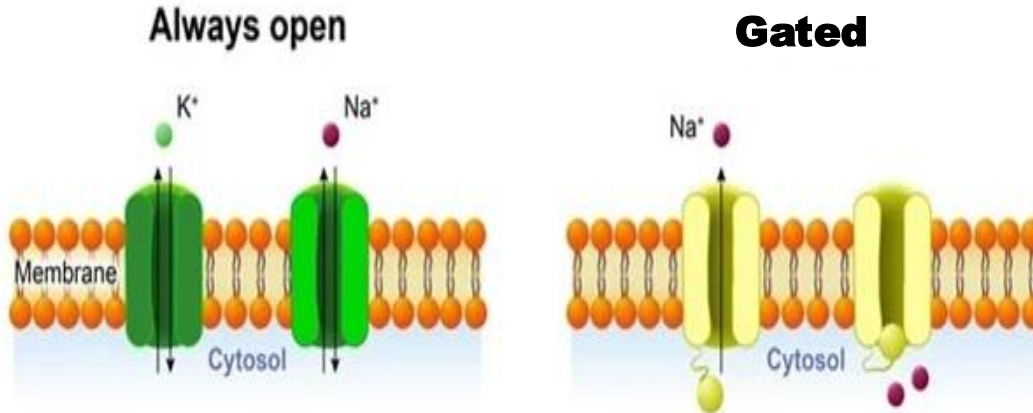
Different methods of transport across the cell membrane

2. Through protein channels:

I. **Leak channels:** Always open (**non-gated**).

II. **Gated:** It's guarded by gate/s on the inner or the outer surface or both.

- The activity of the channels depends on the opening or closure of the gates.
- According to the method of response of the channel's gates, gated-channels are divided into three types:

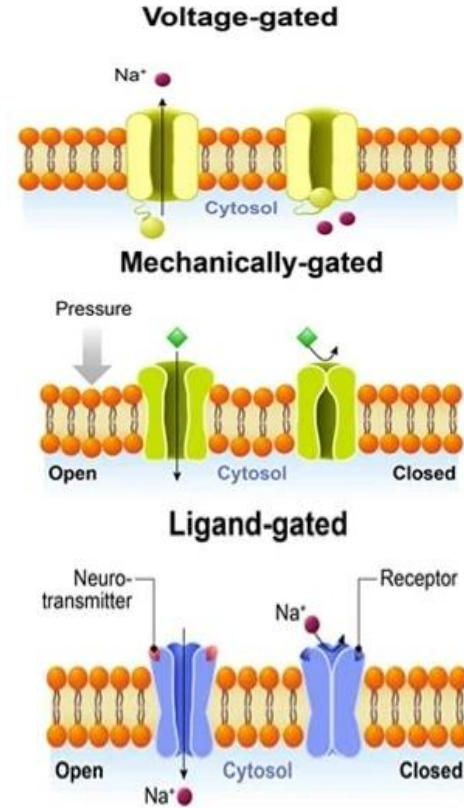


Different methods of transport across the cell membrane

➤ The gated channels are:

1. **Voltage-gated:** i.e., open or close by alterations in the trans-membrane potential.
2. **Mechanically (stretch or pressure) gated:** these mechanosensitive channels play an important role in cell movement.
3. **Ligand-gated:** i.e., open or close when a ligand binds to a specific receptor in the cell membrane. This receptor may be extracellular or intracellular.

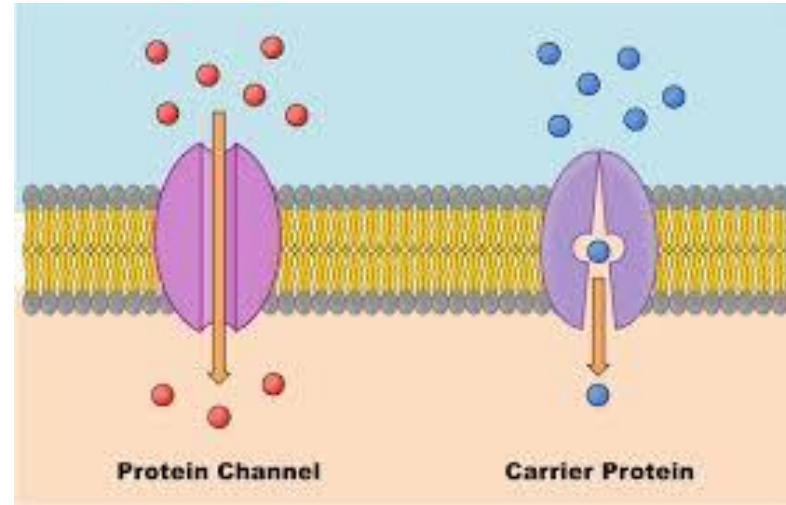
Phosphorylation-gated: where protein phosphorylation or de-phosphorylation regulate opening and closing.



Different methods of transport across the cell membrane

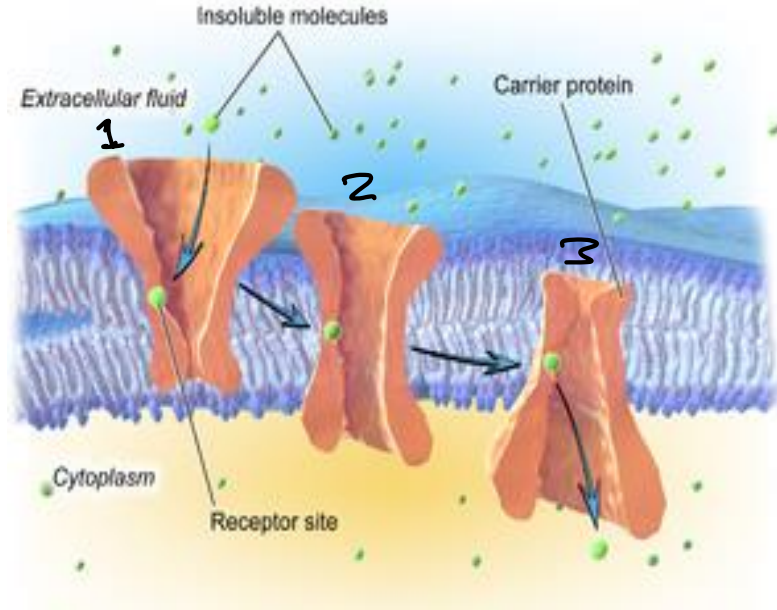
B. Facilitated Diffusion (carrier-mediated).

- They bind **large molecules** (e.g., glucose) from one side of the cell membrane to the other.
- The carrier protein has a channel large enough to transport a specific molecule **part-way** but not all the way through the membrane.



Different methods of transport across the cell membrane

- The molecule to be transported enters the channel and then becomes bound to a **specific receptor** on the protein carrier.
- In a fraction of a second, a **conformational change** occurs in the carrier protein so that the channel now opens to the **opposite side** of the membrane and the attached molecule will be released on the opposite side



Different methods of transport across the cell membrane

➤ In facilitated Diffusion, different carriers differ from each other in:

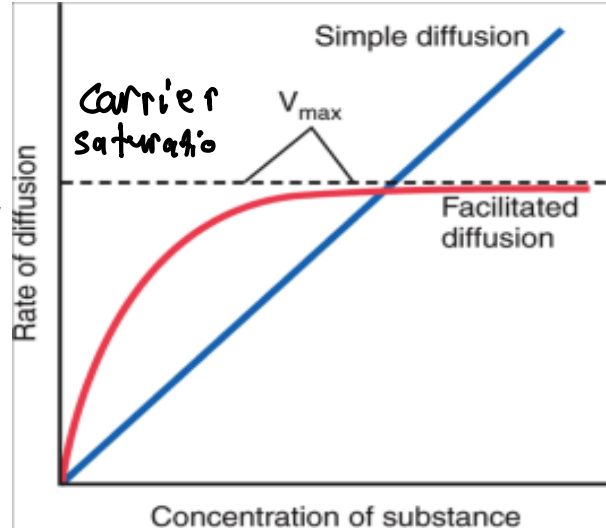
* The **specificity** of the carrier protein.

* The **competition** between **similar** substances (glucose and galactose) for the same carrier.

➤ The **rate** of facilitated diffusion increases proportionately with the concentration gradient **up to** a certain maximum rate (**carrier saturation**).

➤ Its rate can never exceed the rate the carrier protein molecule can undergo change back and forth between its two states.

➤ It is **more sensitive to temperature** than simple diffusion. ★



Thank You!



Conclusion

- The cell membrane is an amphipathic structure.
- The cell membrane contains proteins (55%), phospholipids (25%), cholesterol (13%), other lipids (4%) and carbohydrates (3%).
- Transport through the cell membrane may be by diffusion, active or vesicular transport.
- Diffusion may be simple, facilitated or by osmosis.
- Simple diffusion may be directly through the cell membrane or through protein channels.

Case scenario

A researcher measures the rate at which oxygen moves from the alveolar air into red blood cells across the cell membrane, with no carrier proteins or energy expenditure involved. The rate is found to correlate closely with oxygen's lipid solubility. Which transport mechanism does this describe?

1. Active transport through an ATP-dependent pump
2. Simple diffusion directly through the lipid bilayer
3. Transport through a ligand-gated protein channel
4. Facilitated diffusion via a specific carrier protein

References

Guyton and Hall, 13th edition. Unit V(25); 305-14.

Ganong's review of medical physiology 25th ed. Section I(2); 45-53.

Thank You